दक्षिण बिहार केंद्रीय विश्वविद्यालय

CENTRAL UNIVERSITY OF SOUTH BIHAR

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Question Paper of End-Term Examination Dec 2025

Name of Department	:	Department of Computer Science	Programme	:	MSc. (AI)
Session	:	2025-2027	Semester	:	I
Course Title	:	Artificial Intelligence	Course Code	:	CAI81DC00204
Course Credit	:	4 Max. Marks : 70	Duration	:	3 hours

Enrolment No. (to be filled by Student) : CUSB2502332012

There are three sections (A, B, and C). Answer all questions as per instructions. Points against each question are mentioned in the parenthesis.

Section – A

Answer ALL the questions in approximately 50 words. Each question carries 1 Point.
(10 x 1 = 10 Points)

Q1. Define AI on the basis of "*Thinking Humanly*".

Q2. Write the full form of AO* algorithm.

Q3. What do you understand by *Actuators* in the AI agent?

Q4. Write the heuristics involved in the 8-puzzle problem.

Q5. Which data structure is suitable for DFS algorithm and why?

Q6. What is MAX node in Min-Max tree?

Q7. How propositional logic is different from First order Logic?

Q8. What is Hypothetical Syllogism?

Q9. Draw truth table for the tautology involved in the inference rule Modus Ponens.

Q10. What do you understand by planning problems?

Section – B

Answer ANY FIVE questions in approximately 300 words. Each question carries 6 Points.
(5 x 6 = 30 Points)

Q1. Solve and write the sequence of possible pouring actions in following classic puzzle that can be stated as below:
You are given three unmarked jugs with capacities of exactly 8 liters, 5 liters, and 3 liters. The 8-liter jug is initially full of water, while the 5-liter and 3-liter jugs are empty.
The objective is to measure out exactly **4 liters** of water into one of the jugs.
The only allowed operations are:

- Filling a jug completely from the water source (which is the full 8-liter jug initially).

- Pouring the entire contents of one jug into another, or pouring partially until the destination jug is full or the source jug is empty.
- Water cannot be spilled or measured by any other means (e.g., no additional measuring devices or markings on the jugs are allowed).

Q2. Write down the PEAS (Performance Measure, Environment, Actuators and Sensors) description for the task environment for the following AI agents:

a) *Taxi driving system*

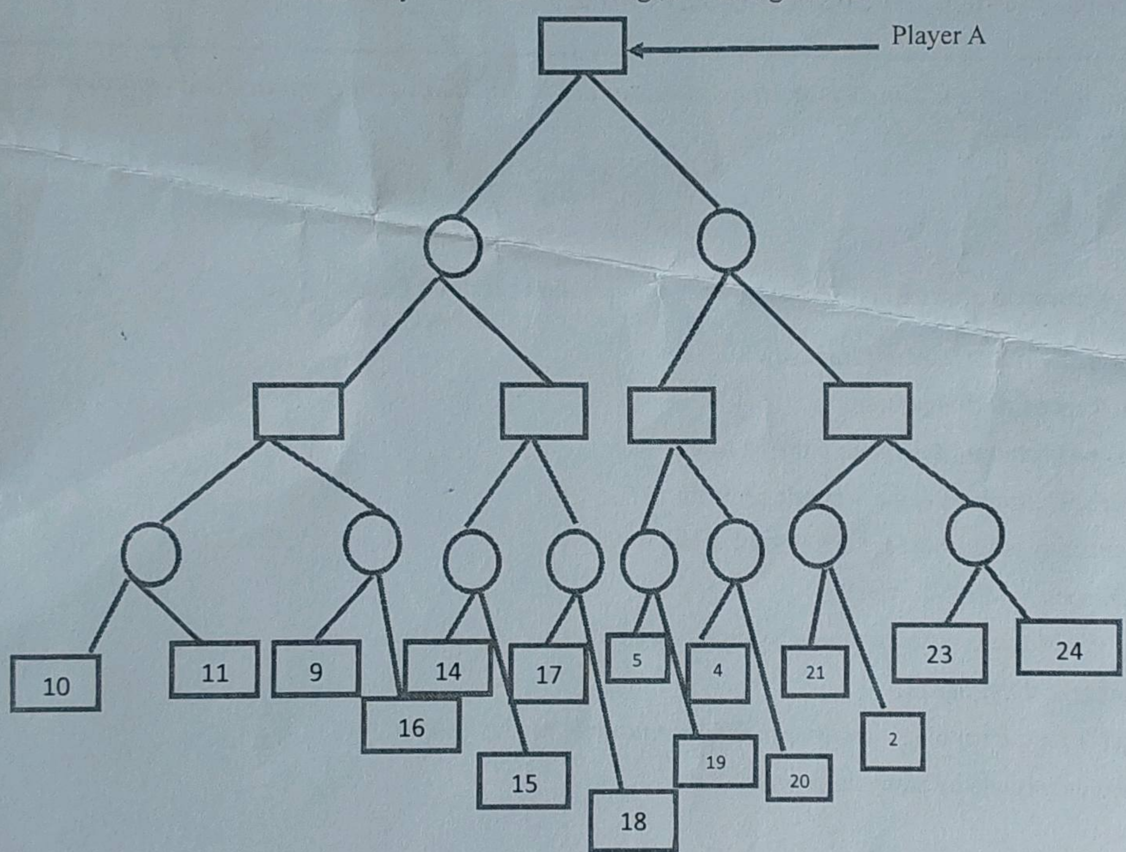
b) *Medical diagnostic system*

c) *Interactive English tutoring system*

Q3. Explain all four letters [S, s, O, G] used in formulating an AI problem with the help of a suitable example.

Q4. Explain the notion of heuristics with the help of a suitable example.

Q5. Show the best move for Player A for the following Min-Max game tree:



Q6. Explain the role of horn clause in the knowledge base.

Section – C

Answer ANY TWO questions in approximately 1000 words. Each question carries 15 Points.
(2 x 15 = 30 Points)

Q1. Discus uniform cost search algorithm assuming each operation have a cost associated with it.

Q2. Solve the following Lewis Carroll's classic example illustrating existential quantification:

Premise 1: "*All lions are fierce.*" (Universal statement)

Premise 2: "*Some lions do not drink coffee.*" (Existential statement)

Conclusion: "*Some fierce creatures do not drink coffee.*" (Existential statement)

Q3. What is a planning agent? What are the assumptions involved in the planning agent? Write the algorithm and pseudo code for simple planning agent. [3+4+8]